Lecture 13

Clifford circuits: simulable, and what they lack

Objectives. *Lecture 12 built codes from commuting Paulis. This lecture asks:* which circuits can a classical computer simulate, and what does a protected computation need beyond them? *Circuits of H, S and CNOT, started on $|0\cdots0\rangle$ and read by Pauli measurements, can be followed through n Pauli strings and simulated classically (Theorem 13.4), although they make entangled states. What they lack is T, which injection gets by consuming the state $|A\rangle = T|+\rangle$ (Theorem 13.6). The threshold theorem (cited) says what a computation needs to survive noise everywhere. This is the first half of the third answer: where the advantage lives.*

By the end of this lecture you will be able to:

- Compute how H, S and CNOT act on Paulis by conjugation, and follow the generators of a stabilizer state through Clifford gates and Pauli measurements, signs included.
- State the Gottesman–Knill theorem with its hypotheses, and sketch its proof.
- Prove that injection applies T, up to a global phase, by consuming $|A\rangle$, and show that $|A\rangle$ is not a stabilizer state.
- State the threshold theorem with its hypotheses.

13.1 The Clifford group

Definition 13.1 (The Clifford group)**.** A gate U on n qubits is a *Clifford* if $UEU^\dagger \in \mathcal{G}_n$ for every $E \in \mathcal{G}_n$. They form the *Clifford group*.

It is a group: conjugation by a Clifford maps the finite set $\mathcal{G}_n$ one-to-one, so onto, itself; so the inverse is a Clifford. Conjugation respects products, and as $Y = iXZ$ every element of $\mathcal{G}_n$ is a phase times a product of X_q's and Z_q's; so the images of the X_q and Z_q decide whether a gate is a Clifford. Multiplying the matrices,

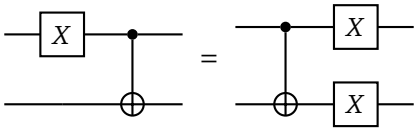
$$HXH^\dagger = Z, \quad HZH^\dagger = X,$$
$$SXS^\dagger = Y, \quad SZS^\dagger = Z,$$

and the CNOT, with qubit 1 its control, maps

$$X\otimes\mathbb{1} \mapsto X\otimes X, \quad \mathbb{1}\otimes X \mapsto \mathbb{1}\otimes X,$$
$$Z\otimes\mathbb{1} \mapsto Z\otimes\mathbb{1}, \quad \mathbb{1}\otimes Z \mapsto Z\otimes Z$$

(Nielsen and Chuang, Fig. 10.7, p. 460). In circuits, an X on the control before a CNOT is

an X on both wires after it,



(13.1)

an equality of matrices; likewise a Z on the target becomes Z on both. So circuits of H, S and CNOT are Cliffords, and so are the Pauli gates, $Z = S^2$ and $X = HS^2H$.

Theorem 13.2 (Generation · *cited*)**.** *Every Clifford on n qubits is, up to a global phase, a circuit of $O(n^2)$ gates H, S and CNOT.*

The count is unconditional (Nielsen and Chuang, Theorem 10.6, pp. 461–462). The phase cannot be dropped: $e^{i\pi/8}\mathbb{1}$ fixes every Pauli, but on one qubit every product of H and S has a power of i as its determinant ($\det H = -1$, $\det S = i$), and $\det(e^{i\pi/8}\mathbb{1}) = e^{i\pi/4}$ is none. And T is no Clifford: $TXT^\dagger = (X+Y)/\sqrt{2}$ is no phase times a Pauli. Only T and $T^\dagger$ of the alphabet (Definition 6.8) lie outside.

13.2 Gottesman–Knill

A gate changes the Paulis that fix a state only by renaming them: if $g\psi = \psi$, then $(UgU^\dagger)U\psi = U\psi$. This is the whole idea.

By Theorem 12.3, a stabilizer group with n independent generators on n qubits fixes a space of dimension $2^0 = 1$: one state, up to a global phase. It is a *stabilizer state*, and its generators, n strings with their signs, determine it. The state $|x\rangle$ is one, with the generators $(-1)^{x_1}Z_1, \ldots, (-1)^{x_n}Z_n$. If ψ is the stabilizer state of $g_1, \ldots, g_n$ and U is a Clifford, then $U\psi$ is the stabilizer state of $Ug_1U^\dagger, \ldots, Ug_nU^\dagger$: conjugation is one-to-one and respects products, so these again commute, are independent and generate a group without $-\mathbb{1}$. Each generator is a string of n letters with a sign, $2n+1$ bits, and a gate on one or two qubits changes, in each generator, the letters on those qubits and perhaps the sign, by the table of §13.1: $O(n)$ steps for all n.

A *measurement* of $g = \pm P$, for a Pauli string $P \neq \mathbb{1}$, is the projective measurement $\frac{1}{2}(\mathbb{1}+g)$, $\frac{1}{2}(\mathbb{1}-g)$; the outcome b is the eigenvalue $(-1)^b$ of g, and the circuit of §12.3 performs it. Reading qubit q is the case $g = Z_q$.

Lemma 13.3 (Measuring a stabilizer state)**.** *Let ψ be the stabilizer state of $g_1, \ldots, g_n$, and measure $g = \pm P$.*

(i) If g anticommutes with a generator, say g_1, replace each other g_j that anticommutes with g by g_1g_j. Each outcome has probability $\frac{1}{2}$, and the outcome b leaves the stabilizer state of $(-1)^b g, g_2, \ldots, g_n$.

(ii) If g commutes with every generator, then $g = (-1)^c h$ for a bit c and a product h of generators, and the measurement gives c with certainty and leaves ψ.

Proof. (i) The new g_j commute with g, as both factors anticommute with it, and the list still consists of independent generators of the same group, as the old g_j is g_1 times the new one. As $g_1\psi = \psi$ and g_1 is self-adjoint,

$$\begin{aligned}\langle\psi|\, g\,|\psi\rangle &= \langle\psi|\, gg_1\,|\psi\rangle \\ &= -\langle\psi|\, g_1g\,|\psi\rangle = -\langle\psi|\, g\,|\psi\rangle\,,\end{aligned}$$

so $\langle\psi|\, g\, |\psi\rangle = 0$, and the outcome b has probability $\langle\psi|\, \frac{1}{2}(\mathbb{1} + (-1)^b g)\, |\psi\rangle = \frac{1}{2}$. It leaves $\psi_b = (\mathbb{1} + (-1)^b g)\psi/\sqrt{2}$, fixed by $(-1)^b g$, as $g^2 = \mathbb{1}$, and by g_j for $j \geq 2$, which commutes with g and fixes ψ. These n commute, and they are independent: $g_2, \ldots, g_n$ are, and g is not $\pm$ a product of them, which would commute with g_1. They fix $\psi_b \neq 0$, so their group does not contain $-\mathbb{1}$.

(ii) As g commutes with S, $g\psi$ is fixed by S, so it lies on the line $C(S)$: $g\psi = \lambda\psi$, with $\lambda = \pm 1$ as $g^2 = \mathbb{1}$. So g is a multiple of $\mathbb{1}$ on $C(S)$, and by Lemma 12.4(iii) $g \in \langle i\mathbb{1}\rangle S$: $g = \mu h$ with $h \in S$, and $\mu = \lambda$, as $h\psi = \psi$. So the measurement gives the c with $(-1)^c = \lambda$, with certainty, and leaves ψ. ∎

To find c, write each string as two rows of n bits, its X-part and its Z-part, Y counting in both. The bits a_j with $h = g_1^{a_1} \cdots g_n^{a_n}$ solve $2n$ linear equations modulo 2, the parts of g's string as sums of the generators' parts, and elimination solves them in $O(n^3)$ steps. Multiplying those generators, phases included, gives h, and comparing it with g gives c. Nielsen and Chuang (p. 463) and de Wolf (p. 176) leave this step out.

Theorem 13.4 (Gottesman–Knill · *sketched*)**.** *Let a circuit on n qubits start in $|0\cdots0\rangle$ and take m steps, each an H, an S, a CNOT or a measurement of a Pauli $\pm P$, where the choice of each step may depend on the earlier outcomes through a classical polynomial-time rule. A classical algorithm with coins, in time polynomial in n and m, gives outcomes with the circuit's joint probabilities, and the generators of the state the circuit leaves.*

Sketch. Start from $Z_1, \ldots, Z_n$. A gate updates the generators by the table, in $O(n)$ steps. A measurement is Lemma 13.3: in case (i) a coin gives b, and the update takes $O(n^2)$ steps, at most n products of strings; case (ii) takes the elimination, $O(n^3)$ steps. After each step the generators are those of the circuit's state, given the outcomes so far, and each outcome has the probability the circuit gives it then; so the outcomes have the circuit's joint probabilities, the products of these. In all, $O(mn^3)$ steps, and the rule's. ∎

The theorem is unconditional. The same algorithm starts from any stabilizer state given by its generators. The Pauli gates are circuits of H and S, and readings of qubits are measurements of Z_q, so both are covered. The algorithm *samples*: it gives outcomes with the right probabilities, the form of de Wolf's Theorem 6 (pp. 175–176), and the theorem says nothing about computing amplitudes. Nielsen and Chuang state it as Theorem 10.7 (p. 464); Gottesman (p. 52) credits Knill.

The input hypothesis matters. Add copies of $|A\rangle$ as inputs, and §13.4 turns each into a T; then the circuits of the theorem reach every circuit over the alphabet of Definition 6.8. A simulation as in the theorem for them would put BQP inside BPP. None is known, and BQP ≠ BPP is a conjecture (Lecture 6).

13.3 The Bell example

Example 13.5 (QQ: the Bell state, by two strings)**.** The circuit (3.6) makes Φ^+ from $|00\rangle$, and its generators go

$$\begin{aligned} & Z_1,\ Z_2 \\ \xrightarrow{H \text{ on } 1} \ & X_1,\ Z_2 \\ \xrightarrow{\text{CNOT}} \ & X_1X_2,\ Z_1Z_2. \end{aligned}$$

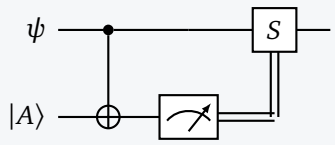
So Φ^+ is the stabilizer state of $X \otimes X$ and $Z \otimes Z$ (Nielsen and Chuang, p. 454; Watrous, 2025, p. 420). Yet each half of it is $\frac{1}{2}\mathbb{1}$ and has no ket (Example 3.11): two strings describe an entangled state exactly. Read qubit 1: Z_1 anticommutes with X_1X_2 and commutes with Z_1Z_2, so by Lemma 13.3(i) each outcome has probability $\frac{1}{2}$, and b leaves the stabilizer state of $(-1)^b Z_1$ and Z_1Z_2, which is $|bb\rangle$, as in Example 3.5. Measure $Y \otimes Y$ instead: it commutes with both generators, and

$$\begin{aligned}(X \otimes X)(Z \otimes Z) &= (XZ) \otimes (XZ)\\ &= (-iY) \otimes (-iY)\\ &= -Y \otimes Y,\end{aligned}$$

so $Y \otimes Y = -(X \otimes X)(Z \otimes Z)$, and by (ii) it gives 1 with certainty, the sign found by multiplying generators.

13.4 Injection

Theorem 13.6 (Injection). *Let ψ be a state of one qubit, the* data*, and $|A\rangle = T|+\rangle = (|0\rangle + e^{i\pi/4}|1\rangle)/\sqrt{2}$. In the circuit*



the reading gives each outcome with probability $\frac{1}{2}$; the outcome 0 leaves the data in $T\psi$, and the outcome 1, after the S, in $e^{i\pi/4}T\psi$. So the data leaves in $T\psi$, up to a global phase.

Proof. Write $\omega = e^{i\pi/4}$ and $\psi = \alpha|0\rangle + \beta|1\rangle$. The CNOT gives

$$\begin{aligned}&\tfrac{1}{\sqrt{2}}\alpha|0\rangle(|0\rangle + \omega|1\rangle)\\ &\quad + \tfrac{1}{\sqrt{2}}\beta|1\rangle(|1\rangle + \omega|0\rangle)\\ &= \tfrac{1}{\sqrt{2}}(\alpha|0\rangle + \omega\beta|1\rangle)|0\rangle\\ &\quad + \tfrac{1}{\sqrt{2}}(\omega\alpha|0\rangle + \beta|1\rangle)|1\rangle.\end{aligned}$$

By Lüders' rule (Definition 3.4), each outcome has probability $\frac{1}{2}(|\alpha|^2 + |\beta|^2) = \frac{1}{2}$. The outcome 0 leaves $\alpha|0\rangle + \omega\beta|1\rangle = T\psi$. The outcome 1 leaves $\omega\alpha|0\rangle + \beta|1\rangle = \omega T^\dagger\psi$, and S makes it $\omega(\alpha|0\rangle + i\bar{\omega}\beta|1\rangle) = \omega T\psi$, as $i\bar{\omega} = \omega$. ∎

The proof is linear in α and β. If the data qubit is entangled with others, α and β become vectors of those qubits, and the same lines give T on the data qubit, each outcome with probability $\frac{1}{2}$, and $e^{i\pi/4}$ a global phase of the whole register. This is the layout of de Wolf's Exercise 6 (p. 181); Aliferis, Gottesman and Preskill draw the same circuit (Fig. 9, p. 28). The classical control of Theorem 13.4 is used: deferring the reading by Theorem 4.5 would need a controlled S, which is no Clifford, as it maps an X on its control outside $\mathcal{G}_2$.

$|A\rangle$ is not a stabilizer state. On one qubit a stabilizer state has one generator, $\pm X$, $\pm Y$ or $\pm Z$ (not $\pm\mathbb{1}$, and a phase $\pm i$ would square to $-\mathbb{1}$), and is fixed by it, so the generator's expectation value is 1. But

$$\begin{aligned}\langle A|X|A\rangle &= \langle A|Y|A\rangle = 1/\sqrt{2},\\ \langle A|Z|A\rangle &= 0,\end{aligned}$$

so these six have the expectation values $\pm 1/\sqrt{2}$ and 0 on $|A\rangle$. The input hypothesis of Theorem 13.4 excludes it, and it supplies what the theorem's circuits lack: with copies of $|A\rangle$, Clifford gates, Pauli measurements and classical control apply T, and $T^\dagger = T^7$.

Names differ. Nielsen and Chuang's $|\Theta\rangle$, de Wolf's $|T\rangle$ and Bravyi and Kitaev's $\left|A_{\pi/4}\right\rangle$ (eq. (2), p. 4) are $|A\rangle$; Bravyi and Kitaev's "T-type" state is another state, and their T (eq. (7), p. 6) is a Clifford. Nielsen and Chuang's Fig. 10.25 (p. 486) reverses the roles: $|A\rangle$ controls, the data qubit is the target and is read, and the correction is SX on the wire of $|A\rangle$, which carries the output. The two corrections do not exchange: with SX above, or S in their circuit, the outcome 1 fails.

13.5 Threshold

Lecture 10 took the encoding, the syndrome circuits and the corrections to be exact. They fail too. Call each preparation of $|0\rangle$, each gate, each qubit resting for one step and each measurement a *location*.

Theorem 13.7 (Threshold · *cited*)**.** *Suppose that*

(a) each location fails independently, with probability at most ϵ; a failure may apply any operation to the location's qubits, and otherwise the location is exact;

(b) gates act in parallel, fresh qubits in $|0\rangle$ can be brought in at every step, the inputs are $|0\rangle$, and the classical processing of outcomes is exact;

(c) fault-tolerant gadgets are given: encoded versions of the locations, with error correction on their blocks, substituted for the locations level by level; if a gadget, with the error corrections before it, contains at most one fault, then decoding its output ideally gives what the ideal location gives on the decoded input.

There is an $\epsilon_0 > 0$, fixed by the gadgets, such that if $\epsilon < \epsilon_0$, then for every $\delta > 0$ every circuit with L locations is simulated by a noisy circuit with $L \cdot \mathrm{polylog}(L/\delta)$ locations, whose output probabilities differ from the ideal ones by at most δ in all.

This is Aliferis, Gottesman and Preskill's Theorem 1 (p. 13; hypotheses pp. 2, 11 and 13–14), unconditional under (a)–(c), with the dependence on δ as Aharonov and Ben-Or state it (Theorem 10, p. 52). The encoding, syndrome circuits and corrections are locations like any other, and below ϵ_0 a computation of any length is still simulated within δ, at a polylogarithmic cost: this answers Lecture 10. A code of distance 3 corrects one error in a block, but one faulty gate can make two; the gadgets of (c) keep one fault harmless, and independence makes two faults in one gadget rare, of order ϵ^2. Even storage needs parallel error correction, and a constant supply of fresh ancillas is necessary (Nielsen and Chuang, p. 494). With noisy classical processing the error stays of order ϵ, however large the codes (Aliferis, Gottesman and Preskill, p. 14); Aharonov and Ben-Or need no classical processing during the computation (p. 1).

Their gadgets for data act on encoded blocks by Clifford operations, and obtain T by this lecture's injection circuit, consuming an encoded $|A\rangle$ that a further gadget prepares (Aliferis, Gottesman and Preskill, Fig. 9 and p. 28): a protected computation needs, beyond Clifford circuits, a supply of magic states.

Values of ϵ_0 depend on the noise model and the gadgets, and are not comparable: 2.73×10^{-5} for Aliferis, Gottesman and Preskill's gadgets under their faults (Theorem 3, p. 38), and about 10^{-6} for Aharonov and Ben-Or's best case, which assumes error-free classical operations (p. 52). None is *the* threshold.

13.6 Where the advantage lives

Clifford circuits make entangled states, the Bell state of Example 13.5 among them, and Theorem 13.4 simulates them all. So entanglement alone does not put a computation beyond a classical computer's reach; in Nielsen and Chuang's words (p. 464), "some quantum computations involving highly entangled states may be simulated efficiently on classical computers". What the simulated circuits lack is a gate outside the Clifford group, T, or the state $|A\rangle$ that injection turns into it. If $\mathsf{BQP} \neq \mathsf{BPP}$, which is a conjecture, the difference needs one of them: this is where the advantage lives (de Wolf, p. 175). Protection does not change this: a protected computation still consumes magic states. Lecture 14 asks whether protection can grow with the size of the system.

Summary.

- *Cliffords:* H, S and CNOT map Paulis to Paulis and generate the Clifford group up to a global phase (cited); T lies outside.
- *Gottesman–Knill* (sketched, unconditional): from $|0\cdots0\rangle$, Clifford gates, Pauli measurements and classical control are simulated by sampling, through n Pauli strings.
- *Injection:* a CNOT, a reading and a classically controlled S apply T, up to a global phase, consuming $|A\rangle$, which is no stabilizer state.
- *Threshold* (cited; unconditional): independent faults below ϵ_0, parallel gates, fresh qubits, exact classical processing and gadgets give computations of any length at a polylogarithmic cost.
- *Next:* Lecture 14, the toric code.

References

Nielsen and Chuang, pp. 454, 459–464, 486 and 494. De Wolf, pp. 174–176 and 181. Watrous (2025), p. 420. D. Gottesman, arXiv:quant-ph/9705052, p. 52. S. Bravyi and A. Kitaev, arXiv:quant-ph/0403025, pp. 4 and 6. P. Aliferis, D. Gottesman and J. Preskill, arXiv:quant-ph/0504218, pp. 2, 11, 13–14, 28 and 38. D. Aharonov and M. Ben-Or, arXiv:quant-ph/9906129, pp. 1 and 52.